How Many Documents Cover a Corpus

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Abstract

A sample drawn from a survey population comes with a margin of error and a formula for the sample size a target precision needs. A sample drawn from a corpus to annotate or to summarize comes with neither. We supply both for one question: what share of the corpus lies within a chosen cosine distance r of some sampled document, in a fixed embedding space. The share that does not is the missing mass at resolution r. Its estimator G needs no labels: it is the share of sampled documents with no other sampled document within r (Maurer, 2022). Under the classical nearest neighbour law, G should depend on sample size n and resolution r only through nr^d, with d the intrinsic dimension of the embedded corpus. We test this on seventeen pairs of corpus and encoder, including forum, census and political social media text. Where the law holds, a fit on a pilot of at most 1,000 documents predicts G at 7,000 documents with an out of sample R^2 of 0.92 to 0.99, and the sample size it prescribes for a target isolation rate falls within a factor of 1.5 of the empirical one in 77 of 83 cases, against errors four to ten times larger for the ecological Chao and Jost extrapolation. For a finite corpus of N documents the missing mass is $G\,(1 - n/N)$, to a mean error of 0.012 at sampling fractions up to 0.7. The law fails on corpora whose local dimension varies, such as slogans beside discussion, and an out of sample R^2 below 0.9 flags them before the sample is drawn.

1 Introduction

Large collections of text are sampled before they are studied. An annotation team codes a few thousand posts out of millions, a language model summarizes a few hundred documents because the corpus does not fit its context, and a topic model is fitted to a slice because the full run is slow. The sampler of a human population has a settled answer to the question such a slice raises. Given the population variance σ^2 and a tolerated error e, the formula $n = z^2\sigma^2/e^2$ gives the sample size, and the estimate comes with a margin of error (Cochran, 1977; Kish, 1965). The sampler of a corpus has no such formula. The sample size is set by budget, and whether the slice covers the corpus is left unasked.

The question has an exact form once a corpus is a set of points. Embed each document, choose a distance r that says what counts as the same content, and ask what share of the corpus lies farther than r from every sampled document. We call that share the missing mass at resolution r. In the discrete case, where two documents either share a category or do not, it is the quantity Good and Turing estimated from the share of draws whose category was seen once (Good, 1953). Ecologists built a sampling theory on that estimate: a rarefaction curve gives the coverage a smaller sample would have had, an extrapolation curve gives the coverage a larger sample will have, and together they say how many more units a target coverage needs (Chao and Jost, 2012). Maurer (2022) carried the estimator into metric spaces. His statistic G is the share of sampled points with no other sampled point within r, and he proved bounds on its bias and variance. The estimator has had no application to text, and Maurer's treatment says nothing about how G falls as the sample grows.

How G falls follows from a classical result. For points drawn from a density on a d-dimensional set, the chance that a point has no neighbour within r among n others decays like $\exp(-c\,n\,r^d)$, the law behind nearest neighbour estimators of intrinsic dimension (Levina and Bickel, 2005; Facco et al., 2017) and behind the isolated vertices of a random geometric graph (Penrose, 2003). We take this law, which has not been used to plan samples, and test whether it holds well enough on embedded text to set a sample size. Three results carry the paper.

1. On corpora where the law holds, G depends

on n and r through nr^d, with d the intrinsic dimension estimated from a pilot of 2,000 documents and two shape constants fitted on the pilot. A fit on samples of at most 1,000 documents predicts G at 7,000 documents with an out of sample R^2 of 0.92 to 0.99. The sample size it prescribes for a target isolation rate falls within a factor of 1.5 of the empirical sample size in 77 of 83 cases across eleven such corpora and two encoders. A per radius Chao and Jost extrapolation, the standard ecological tool, errs four to ten times more.

2. For a finite corpus of N documents the missing mass is $G\,(1-n/N)$, because the sampled documents are covered by definition. The correction brings the mean error from 0.11 to 0.012 over sampling fractions up to 0.7.

3. The law fails on corpora whose local dimension varies, such as political posts where slogans of dimension 2 sit beside discussion of dimension 16. An out of sample R^2 below 0.9 flags these corpora, and on them the formula's predictions fall within a factor of 1.5 in only 10 of 23 cases.

In practice the procedure is short. A user picks the resolution r that matches the question, embeds a pilot, estimates d and the two constants, checks the fit, and takes the sample size from the formula. After the sample is drawn, $G\,(1 - n/N)$ reports the share of the corpus it leaves uncovered.

2 Background

2.1 Sample coverage in the discrete case

A sample of n draws from a population of categories covers the categories it has seen. The mass it has missed is the total probability of the unseen categories. Let f_k be the number of categories that appeared exactly k times in the sample. Good (1953) showed that f_1/n, the share of draws whose category appeared once, estimates the missing mass with small bias. Ecologists took the estimator as the basis of a sampling theory for species. Chao (1984) bounded the number of unseen species from f_1 and f_2, and Chao and Jost (2012) wrote the expected coverage of a sample enlarged by m draws as a closed form in n, f_1 and f_2, so that a survey of a habitat can be stopped at a target coverage. Efron and Thisted (1976) asked how many words Shakespeare knew with the same machinery, and Orlitsky et al. (2016) gave the optimal prediction of the number of unseen species for a sample up to about $n \log n$ times larger than the one in hand.

The estimators have reached text and machine learning a few times. Hajič and Moss (2025) apply the Chao bound to a catalogue of chant to say how far a collection is from complete. Tran et al. (2017) fit species models to the answers to an open ended survey question to predict when new answers stop appearing. Chung et al. (2018) count the unknown examples of a training set with the same estimator, a species being a unique example, and Pal et al. (2026) estimate the mass of deployment states a classifier has never seen. In all four a species is a discrete label and two items either match or do not.

2.2 The missing mass in a metric space

Maurer (2022) removed the discreteness. In a metric space with distance $d(\cdot,\cdot)$, the missing mass of a sample $X_1, \dots, X_n$ at scale r is the probability that a new draw lies farther than r from every X_i. Maurer calls it the conditional missing mass because it is conditional on the sample drawn. His estimator is

$$G(X, r) = \frac{1}{n}\,\big|\{k : d(X_k, X_i) > r \text{ for all } i \neq k\}\big|, \tag{1}$$

the share of sample points with no other sample point within r. In the discrete case, with $d(x, y) = 1$ for $x \neq y$ and $r < 1$, it is f_1/n. Maurer proved that G exceeds the conditional missing mass by at most $1/n$ in expectation and that the variance of G minus the conditional missing mass is at most $3/n$ (his Theorem 2.1). The paper is theory. It sketches anomaly detection and coding as applications, and it contains no data, no embeddings and no statement about how G falls with n. That dependence is the subject of the nearest neighbour distance law: for a density f on a set of dimension d, the probability that a point has no neighbour within r among n others is close to $\exp(-n f(x)\, v_d\, r^d)$, with v_d the volume of the unit ball (Evans et al., 2002; Penrose, 2003). The same law underlies the correlation dimension of Grassberger and Procaccia (1983) and the estimators of Levina and Bickel (2005) and Facco et al. (2017). The nearest ecological construction is the attribute diversity of Chao et al. (2019), in which two species closer than a threshold τ count as partly the same and diversity is reported as a function of τ. We treat r the same way, as a resolution the user chooses.

2.3 Diversity of a sample in embedding space

Several measures describe how spread out a set of embedded points is. The Vendi score is the exponential of the entropy of the eigenvalues of the sample's cosine kernel matrix, an effective number of distinct items (Friedman and Dieng, 2023), and its cousins extend it to the orders of a Hill number (Pasarkar and Dieng, 2024). The precision and recall of Kynkäänniemi et al. (2019) count generated points inside nearest neighbour balls around a reference set, the density and coverage metrics of Naeem et al. (2020) make those counts robust to the reference set's own spread, and MAUVE compares two distributions on a quantized embedding space (Pillutla et al., 2021). Each describes the sample in hand or compares two samples. None asks what a larger sample would cover, and none has a parameter in the role of a population variance. Core-set selection chooses points to minimise the largest distance from any point to the chosen set (Sener and Savarese, 2018), and semantic deduplication removes points within a small distance of another (Abbas et al., 2023; Lee et al., 2022). Both need the full embedded corpus, and neither says how many points a target needs. The intrinsic dimension of learned representations has been measured for images (Pope et al., 2021), for the layers of deep networks (Ansuini et al., 2019) and for text, where it separates human from generated writing (Tulchinskii et al., 2023). We use it as the corpus parameter that sets a sample size.

2.4 Saturation in practice

Where a sample size for text is set at all, it is set by a stopping rule. Qualitative researchers stop interviewing when new interviews add no new code. Guest et al. (2006) found that point at twelve interviews, and a review of 23 empirical tests by Hennink and Kaiser (2022) puts it at 9 to 17. De Paoli and Mathis (2024) carry the rule to coding by a language model and track the slope of the count of unique codes. TopicGPT reports that its topic count levels off after about 600 documents and stops when 200 documents add no topic (Pham et al., 2024). Maeda et al. (2025) fit an LDA model with eight topics to 522 open ended survey responses and find that a sample of 100 responses recovers over 60% of the topics found on the full corpus and 250 recover about 70%. These are observations of a coverage curve. The present paper gives the curve a parameter, the intrinsic dimension d, and an estimator.

3 Method

3.1 Setting and estimand

A corpus is a finite set of N documents $x_1, \dots, x_N$. Each is embedded by a fixed encoder and normalised, and the distance between two documents is the cosine distance of their embeddings. A sample S is a set of n documents drawn without replacement. The user chooses a resolution r: two documents within r of each other count as the same content for the purpose at hand. The quantity to estimate, for the sample actually drawn, is the missing mass at resolution r,

$$M(S, r) = \frac{1}{N} \left| \{ j : d(x_j, x_i) > r \text{ for all } i \in S \} \right|, \tag{2}$$

the share of corpus documents with no sampled document within r. Coverage is $1 - M$. The estimator is Maurer's G of Equation 1, computed on the sample alone from each sampled document's distance to its nearest sampled neighbour. We write $G(n, r)$ for its value on a sample of size n.

The estimand is the finite corpus in hand. Sampling exists because the corpus is too large to annotate or to place in a model's context, and the full corpus is available to check the estimate.

3.2 The collapse law

Suppose the embedded documents lie near a set of intrinsic dimension d. The mass of a ball of radius r around document x is then about $c(x)\, r^d$, where $c(x)$ is proportional to the local density, and the probability that none of n sampled documents falls in the ball is about $\exp(-c(x)\, n\, r^d)$. Averaging over documents gives

$$\mathbb{E}\, G(n, r) \approx \mathbb{E}_x \left[\exp(-c(x)\, n\, r^d) \right] = \mathcal{L}_c(n\, r^d), \tag{3}$$

the Laplace transform of the distribution of c evaluated at nr^d. Whatever the density, G depends on n and r through the product nr^d alone, so long as one d holds across the radii of interest. The shape of the curve depends on how c varies: a constant c gives a pure exponential, and a spread of c gives a curve that falls more slowly. We describe the shape by a stretched exponential, $\exp(-(\kappa\, n\, r^d)^b)$ with $b \le 1$, which is the Laplace transform of a particular mixing distribution, and fit

$$\log\big(-\log G(n, r)\big) = a + b \log\big(n\, r^d\big), \tag{4}$$

with d estimated beforehand and a and b fitted by least squares. The grid has eleven sample sizes from 100 to 7,000 (capped at $0.7N$), twelve radii spaced geometrically from the 5th to the 99th percentile of a pilot's nearest neighbour distances, and five independent draws per sample size. Only cells with $0.02 < G < 0.98$ enter the fit. The law fails when the dimension itself changes with the radius, because no single d then aligns the curves for different r.

The dimension d comes from the TwoNN estimator of Facco et al. (2017), which uses the ratio of each point's second to first nearest neighbour distance, averaged over five pilots of 2,000 documents. As a check that owes nothing to TwoNN, we also regress $\log(-\log G)$ on $\log n$ and $\log r$ separately. If the law holds, the coefficient on $\log r$ is $b\,d$ and the coefficient on $\log n$ is b, so their ratio estimates d.

Solving Equation 4 for n gives the sample size at which the isolation rate at resolution r reaches a target ε,

$$n^*(\varepsilon, r) = \left(\frac{\log(1/\varepsilon)}{e^a}\right)^{1/b} r^{-d}. \qquad (5)$$

The corpus enters through three numbers. The dimension d sets how fast the required sample grows as the resolution becomes finer: each halving of r multiplies n^* by 2^d. The constants a and b set the level and are fitted on the pilot. We report a 90% interval for n^* by resampling the draws used in the fit.

3.3 Finite population correction

G estimates the missing mass of the distribution that generated the corpus. When the estimand is the corpus itself, the n sampled documents have distance zero to the sample, and each of the $N - n$ unsampled documents is isolated from the sample with probability close to G. The expected missing mass of the corpus is therefore

$$\hat{M}(n, r) = G(n, r)\,\big(1 - n/N\big). \qquad (6)$$

The factor is the one survey sampling calls the finite population correction. There it scales the variance of a mean, and here it scales an expectation, because sampled units are covered by definition.

3.4 A diagnostic for corpora with more than one scale

The law assumes one dimension across the radii of the grid. We fit Equation 4 on cells with $n \leq 1{,}000$ and compute R^2 on cells with $n > 1{,}000$. We call a corpus collapsing when this out of sample R^2 is at least 0.9. The threshold was chosen on the corpora of this paper and is reported with the error rates on both sides of it. When the law fails, two probes locate the cause: the share of documents with a near duplicate (a nearest neighbour within cosine distance 0.02), and the spread of the TwoNN dimension across twenty k-means clusters of the corpus.

4 Data

Table 1 lists the corpora with their results. Two synthetic families have a known answer. Gaussian manifolds place 20,000 points on a random d-dimensional subspace of $\mathbb{R}^{64}$ for $d \in \{2, 4, 8\}$, so the true intrinsic dimension is known. A Zipf topic mixture draws 20,000 documents from 200 topic centres with Zipf weights, isotropic Gaussian noise and normalisation to the unit sphere, so the topic of every document is known.

Ten real corpora come from four sources. Three forum corpora were assembled for earlier work on corpus summarization: AskDocs (medical questions from Reddit), Trump (posts about the 2020 election), and TIFU (stories told in the first person on Reddit), 10,000 documents each after cleaning and deduplication. They are embedded twice, with GTR-T5-base (Ni et al., 2022) at three random samples of the source (seeds 42 to 44) and with Qwen3-Embedding-8B (Zhang et al., 2025) at seed 42. Five corpora are keyword slices of a 2020 export of tweets about the United States Census: random, citizens, trump, covid and illegal, 7,610 to 30,136 tweets after URL and mention removal and deduplication of identical text. Two corpora are uniform random samples of 50,000 English posts each, drawn from collections of Truth Social and Bluesky posts between May and December 2024 and deduplicated on text, which also removes reposts that copy their original. The Truth Social collection holds the threads under posts about the 2024 candidates, and the Bluesky collection is a broad pool of political posts, so neither is a sample of its platform. All census, Truth Social and Bluesky corpora are embedded with Qwen3-Embedding-8B at 4,096 dimensions, truncated at 256 tokens and normalised. Cosine distance is used throughout. It is not a metric, and $\sqrt{2(1 - \cos)}$, the Euclidean distance between normalised vectors, gives the same nearest neighbours.

The posts are public, and the analysis uses only their embeddings and corpus level statistics. No post or account is quoted or identified in the paper.

5 Results

5.1 The estimator tracks true topic coverage

The Zipf mixtures give every document a topic, so the true topic mass covered by a sample is known. Over four settings (50 to 500 topics, two noise levels), three seeds and eleven sample sizes from 50 to 5,000, coverage $1 - G$ at a radius of half the median pairwise distance of a pilot tracks the true coverage with Spearman $\rho = 0.968$ and a mean absolute error of 0.029 (132 samples). Good-Turing on the true labels, which G does not see, reaches $\rho = 0.989$ and error 0.016. At a tenth of the median distance the agreement is gone ($\rho = 0.31$). The radius matters, and the collapse law gives the dependence on it.

5.2 The isolation rate collapses on nr^d

Table 1 reports the fit of Equation 4 with d fixed at its TwoNN estimate. Eleven corpora pass the diagnostic: the three manifolds, AskDocs and TIFU under both encoders, the random census slice, Truth Social and Bluesky. On them the fit on cells with $n \leq 1{,}000$ explains 92% to 99% of the variance of $\log(-\log G)$ on cells with n up to 7,000, and the G it predicts at the largest n has a mean absolute error of 0.015 to 0.042. Fitting on $n \leq 300$ instead, a 23-fold extrapolation, leaves the out of sample R^2 between 0.92 and 0.99. Across the three GTR samples of AskDocs and TIFU the out of sample R^2 varies by at most 0.012. Figure 1 shows the surfaces against nr^d.

The fit with $\log n$ and $\log r$ as separate regressors recovers the dimension without TwoNN. The ratio of the two coefficients is 2.1, 4.4 and 7.3 on the manifolds of true dimension 2, 4 and 8. The exponent b is below 1 on every corpus: 0.62 to 0.80 on the manifolds, 0.49 to 0.54 on AskDocs and TIFU, 0.43 on Truth Social and Bluesky, and 0.25 to 0.30 on the census slices.

The per radius Chao and Jost extrapolation is the ecological standard for the same task (Chao and Jost, 2012). From the singletons and doubletons at $n = 1{,}000$ at each radius it predicts G at the largest n with a mean absolute error of 0.10 to 0.22 on the corpora that pass the diagnostic, four to ten times the error of the collapse law. Its premise of a discrete species does not hold at a fixed radius, where a document's neighbourhood overlaps those of others.

5.3 The sample size formula

For every corpus, Equation 4 was fitted on cells with $n \leq 1{,}000$, Equation 5 gave n^* for targets $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the mean surface crosses ε beyond $n = 1{,}000$, and the prediction was compared with the crossing interpolated on the log scale between grid points. On the corpora that pass the diagnostic, 77 of 83 predictions fall within a factor of 1.5 of the empirical sample size and 81 of 83 within a factor of 2, with median ratios of 0.74 to 1.30 per corpus. On the corpora that fail it, 10 of 23 fall within 1.5 and 14 of 23 within 2, and the Zipf mixture is off by a factor of 70. Appendix B lists every corpus. The 90% interval from resampling draws is narrow, a median of 7% of n^*, and covers the empirical crossing in only 23 of 83 cases, so the error of the formula comes from the shape of the law and the interval reflects only sampling noise in the pilot.

A random sample can be compared with a sample chosen for coverage. Greedy farthest point selection, the core-set construction of Sener and Savarese (2018), minimises the largest distance from any document to the chosen set. To bring the missing mass at a radius below a target, a random sample needs 0.6 to 1.0 times as many documents as greedy selection on the forum and census corpora (median 0.62 over 76 cases), and 0.12 times as many on Truth Social and Bluesky. Greedy selection spends its picks on outlying documents, which are few, while the missing mass weighs documents equally. For the share of the corpus left uncovered, random sampling is the better design, and it needs no embedding of the full corpus.

5.4 Where the collapse fails

Seven corpus and encoder pairs fail the diagnostic: the Trump forum corpus under both encoders, four of the five census slices, and the Zipf mixture. The diagnostic is conservative: on three of them (census covid and illegal, Trump under Qwen3) the formula still predicts within a factor of 1.5 in 7 of 8 cases, while on census citizens, Trump under GTR and the Zipf mixture it fails in 11 of 13. Two probes locate the difference on the GTR embeddings. In a 5,000-document sample of Trump, 3.4% of documents have a near duplicate within cosine distance 0.02, against 0.0% in AskDocs and TIFU and below 0.1% on Truth Social and Bluesky.

Corpus	Encoder	N	d (sd)	b	R^2 in	R^2 out	error, collapse	error, Chao–Jost
Manifold d=2	none	20,000	1.99 (0.05)	0.80	0.963	0.960	0.027	0.105
Manifold d=4	none	20,000	4.01 (0.16)	0.72	0.950	0.922	0.036	0.130
Manifold d=8	none	20,000	7.98 (0.21)	0.62	0.961	0.950	0.042	0.151
AskDocs	GTR	10,000	13.6 (0.36)	0.51	0.961	0.949	0.042	0.183
AskDocs	Qwen3	10,000	12.8 (0.39)	0.54	0.979	0.991	0.015	0.205
TIFU	GTR	10,000	18.3 (0.33)	0.50	0.961	0.955	0.040	0.162
TIFU	Qwen3	10,000	13.6 (0.35)	0.53	0.970	0.959	0.039	0.182
Census random	Qwen3	7,610	12.9 (0.39)	0.27	0.974	0.966	0.040	0.167
Truth Social	Qwen3	50,000	12.6 (0.15)	0.43	0.976	0.977	0.025	0.195
Bluesky	Qwen3	50,000	16.0 (0.69)	0.42	0.969	0.957	0.037	0.215
Census illegal	Qwen3	17,202	11.4 (0.33)	0.26	0.906	0.852	0.026	0.202
Census covid	Qwen3	17,837	10.7 (0.40)	0.30	0.891	0.791	0.045	0.206
Trump	Qwen3	10,000	9.0 (0.51)	0.26	0.875	0.782	0.046	0.243
Census trump	Qwen3	30,136	9.0 (0.84)	0.26	0.891	0.781	0.043	0.237
Census citizens	Qwen3	7,849	9.1 (0.24)	0.27	0.837	0.734	0.043	0.156
Trump	GTR	10,000	9.0 (0.48)	0.27	0.802	0.626	0.054	0.257
Zipf, 200 topics	none	20,000	11.3 (0.35)	0.21	0.671	0.618	0.136	0.096

Table 1: The collapse law on 17 corpus and encoder pairs, sorted by out of sample R^2; the line separates the pairs that pass the diagnostic (out of sample $R^2 \geq 0.9$). d is the TwoNN dimension, mean and sd over five pilots of 2,000 documents (and over three samples for the GTR forum corpora). b and both R^2 come from the fit of Equation 4 on cells with $n \leq 1{,}000$; R^2 out is computed on cells with $n > 1{,}000$. The errors are mean absolute errors of the predicted G at the largest n (7,000, or $0.7N$), for the collapse law and for a per radius Chao and Jost extrapolation from $n = 1{,}000$.

The TwoNN dimension of Trump's twenty k-means clusters runs from 2.0 to 16.2 (coefficient of variation 0.52), against 0.15 for AskDocs, 0.10 for TIFU, 0.16 for Truth Social and 0.13 for Bluesky. The clusters of dimension near 2 hold slogans and the clusters near 16 hold discussion. Removing the 266 documents that have a near duplicate raises the out of sample R^2 only to 0.74.

Section 3.2 says why a spread of local dimension breaks the law and a spread of density does not. A spread of density changes the shape of the curve in nr^d, which b absorbs. A spread of dimension makes the effective d change with the radius: at small r the slogan clusters, where documents sit on a nearly flat set, dominate the isolation count, and at large r the discussion clusters do. No single d aligns the curves for different radii, which is the separation of colours visible for Trump in Figure 1. The Zipf mixture shows the same pattern for another reason. At small radii its documents look like noise around their centres, with the dimension of the noise, and at large radii they look like 200 points, the centres.

The failing census slices are those selected by a keyword that recurs in slogans and calls to action (citizens, trump, covid, illegal). The random slice, which was not selected on a keyword, passes.

n/N	M (true)	G	$G\,(1-n/N)$	error
0.01	0.762	0.768	0.760	0.032
0.05	0.590	0.623	0.592	0.014
0.10	0.497	0.547	0.492	0.010
0.25	0.347	0.459	0.344	0.006
0.50	0.198	0.396	0.198	0.004
0.70	0.110	0.366	0.110	0.003

Table 2: Missing mass of the finite corpus by sampling fraction, mean over eight Qwen3 corpora and three radii (the 25th, 50th and 75th percentiles of the pilot's nearest neighbour distances). The last column is the mean absolute error of the corrected estimate.

5.5 The finite population correction

Table 2 compares G and $G\,(1-n/N)$ with the true missing mass M of the finite corpus on the eight forum and census corpora under Qwen3, at sampling fractions from 0.01 to 0.7 and three radii, 144 cells. Uncorrected, G has a mean absolute error of 0.114 and a maximum of 0.49. Corrected, the mean error is 0.012 and the maximum 0.11, and at $n/N = 0.5$ the error is 0.004. The same comparison on the three GTR forum corpora and the Zipf mixture, at n from 100 to 5,000, gives 0.068 uncorrected and 0.015 corrected. The correction holds on the corpora that fail the collapse.

5.6 Choosing the resolution

[Pending: calibration of r against model judgments of same content on document pairs binned by co-

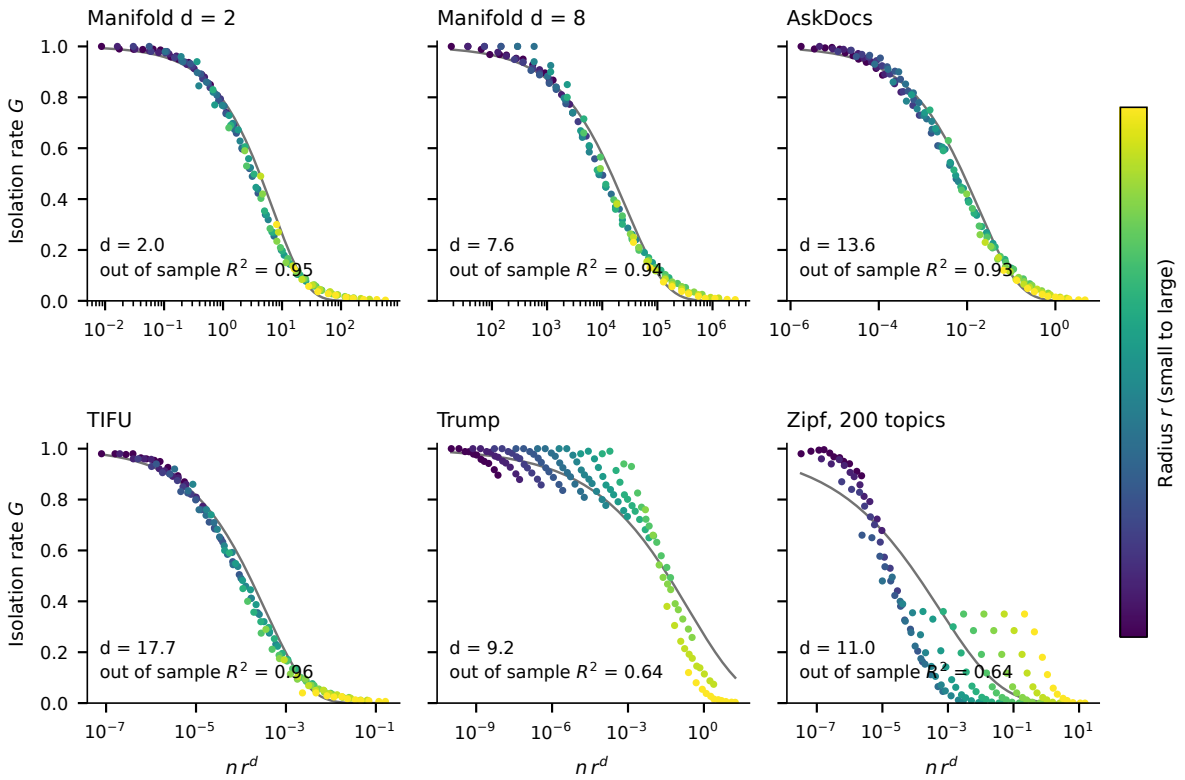


Figure 1: Isolation rate G against nr^d on a log scale, one panel per corpus (GTR encoder for the forum corpora, first sample), colour by radius r. Points from different radii fall on one curve where the law holds (the manifolds, AskDocs, TIFU) and separate where it fails (Trump, the Zipf mixture). The grey line is the fit of Equation 4 on all cells.

sine distance.]

6 Discussion

The procedure a user follows has four steps. Choose the resolution r that matches the question, using the calibration of Section 5.2 as a guide. Embed a pilot of about 2,000 documents, estimate d by TwoNN, and fit Equation 4 on subsamples of the pilot. If the out of sample R^2 is at least 0.9, take n^* from Equation 5. If it is lower, the corpus has more than one scale, and the probes of Section 5.4 say where. After the sample is drawn, report $G\,(1 - n/N)$ as the share of the corpus the sample leaves uncovered. No step needs labels or a GPU once the embeddings exist.

The survey analogy holds in one place and is instructive in two. The factor $1 - n/N$ is the same as the survey correction, applied here to an expectation. The dimension d plays part of the role of the variance: it is the corpus property that sets how fast the required sample grows as the target becomes finer, and like the variance it is estimated from a pilot, here with two shape constants alongside it. The margin of error is the third place. A survey's margin bounds the error of a proportion, and a corpus sample's plain mean already does that job, to within 0.01 for cluster proportions at $n = 1{,}000$ (Appendix E). The missing mass bounds the mass of content the sample has no example of, which is the error that matters when the task is to list what is in the corpus. A report of a corpus sample can carry both numbers.

Coverage of a list of themes is a different quantity. When themes are a partition of the corpus with masses $m_1, \ldots, m_K$, the expected share of theme mass present in a sample of n is $\sum_k m_k(1 - (1 - m_k)^n)$, set by the masses alone. With inventories of 10, 30 and 100 themes per corpus and with inventories of skewed mass, this expression matched the observed coverage of 1,200 samples from eight corpora to a mean error of 0.004, and the curves for a given inventory size nearly coincided across corpora (Appendix C). A theme inventory therefore measures coverage relative to its own granularity, and two corpora that differ tenfold in breadth at a fixed r look the same through it. The stability of

a free summary written from a sample is relative in the same way (Appendix D). Breadth at an absolute scale is visible only to a measure at that scale, which is what G provides.

The diagnostic has a use beyond the formula. A corpus that fails it holds material at more than one scale, and a single sample size will oversample one part and undersample the other. The probe of local dimension across clusters locates the parts, and a sample stratified on them is the natural design. We have not tested it.

Limitations

The resolution r is a cosine distance in one encoder's space. The law held under both encoders on AskDocs and TIFU, but d moved with the encoder (18.3 under GTR against 13.6 under Qwen3 on TIFU), so a user must fix the encoder before choosing r and report the two together.

The required sample size grows as r^{-d}, so the formula is useful only at resolutions where n^* is a feasible number. At $d = 15$, halving r multiplies n^* by about 33,000. The formula plans samples at the resolution of the question and cannot make a fine resolution affordable.

The constants a and b are fitted on the pilot. We have an account of why b is below 1 but no expression that predicts it.

The diagnostic threshold of 0.9 was chosen on the corpora of this paper. The corpora are English social media and forum text, and the Truth Social and Bluesky collections are political pools, so their d describes those collections.

The test unit is the document. Posts by the same account are correlated, and a design that samples accounts would need a cluster version of the estimator.

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A Synthetic smoke test

Zipf topic mixtures with 20,000 documents in 64 dimensions, four settings (50 topics with Zipf exponent 0.5; 200 and 500 topics with exponent 1.0;

200 topics with noise 1.0), three seeds, samples at eleven sizes from 50 to 5,000. Coverage $1 - G$ at radius α times the median pilot pairwise distance, against the true topic mass coverage: $\alpha = 0.5$ gives $\rho = 0.968$ and error 0.029; $\alpha = 0.4$ gives 0.797 and 0.091; $\alpha = 0.3$ gives 0.677 and 0.209; $\alpha \leq 0.2$ fails.

B The sample size formula by corpus

Equation 4 was fitted on cells with $n \leq 1{,}000$, five draws per cell. Equation 5 then gave n^* for $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the mean surface crosses ε at some $n > 1{,}000$, and the crossing was interpolated on the log scale between grid points. Table 3 reports the ratio of predicted to empirical sample size. The grid steps by a factor of about 1.5 between adjacent sample sizes, and the interpolated crossing varies across the five draws by a median of 6%.

Corpus	Enc.	cases	median	≤ 1.5	≤ 2
AskDocs	Qwen3	5	1.04	5	5
Truth Social	Qwen3	6	0.88	5	6
Census random	Qwen3	3	0.81	2	2
Manifold d=2	none	9	1.15	9	9
TIFU	Qwen3	6	1.08	6	6
Bluesky	Qwen3	5	0.74	3	4
TIFU	GTR	17	1.24	17	17
Manifold d=8	none	7	1.12	7	7
AskDocs	GTR	16	0.96	14	16
Manifold d=4	none	9	1.09	9	9
Census illegal	Qwen3	3	1.24	2	3
Census covid	Qwen3	3	1.22	3	3
Trump	Qwen3	2	1.14	2	2
Census trump	Qwen3	2	1.56	1	2
Census citizens	Qwen3	2	2.06	0	1
Trump	GTR	6	1.86	2	3
Zipf, 200 topics	none	5	69	0	0

Table 3: Predicted over empirical sample size for a target isolation rate, in the order of Table 1. GTR rows pool three samples of the forum corpus.

C Coverage of a theme inventory

Each of the eight forum and census corpora received inventories of themes by k-means on the full Qwen3 embeddings at $K = 10$, 30 and 100, and two inventories with skewed masses at $K = 30$: one from k-means++ seeding without refinement, one by merging the 100 clusters into 30 groups of Zipf sizes. The effective number of themes (the exponential of the entropy of the masses) ranges from 6.8 to 91. For each inventory and the 1,200 samples of Section D, Table 4 gives the range across corpora of the share of theme mass present. The expected coverage $\sum_k m_k(1 - (1 - m_k)^n)$ matches the observed value to a mean absolute error of 0.004 over all cells. For the $K = 30$ inventory, Gemma 4 (Gemma Team, 2026) was shown each sample and the list of named themes and asked which were present, with a supporting document for each. The mass of themes it reported rose with n along the label coverage curve in every corpus, and the share of its reports that the labels confirm rose from 0.73 at $n = 25$ to 0.98 at $n = 400$.

Inventory	n=25	n=50	n=100
k-means, K=10	0.89 to 0.94	0.98 to 1.00	1.00
k-means, K=30	0.58 to 0.64	0.82 to 0.85	0.94 to 0.98
k-means, K=100	0.25 to 0.30	0.44 to 0.48	0.67 to 0.72
seeding only, K=30	0.60 to 0.72	0.82 to 0.88	0.94 to 0.96
Zipf merge, K=30	0.79 to 0.84	0.87 to 0.91	0.93 to 0.96

Table 4: Share of theme mass present in a sample, range across eight corpora.

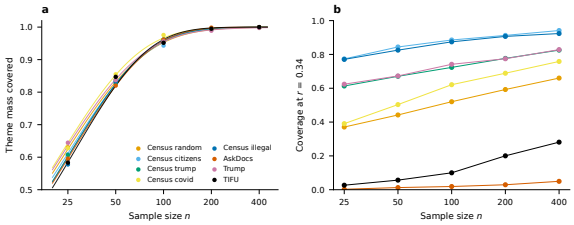


Figure 2: (a) Share of theme mass present in a sample against n for the $K = 30$ inventory, eight corpora (points), with the expected coverage from each corpus's masses (lines). (b) Coverage $1 - G$ at the fixed radius $r = 0.34$ for the same samples.

D Summary stability

For each of the eight forum and census corpora and each $n \in \{25, 50, 100, 200, 400\}$, thirty independent samples were summarized by Gemma 4 at temperature 0 in one paragraph of at most 150 words, and the summaries were embedded with Qwen3-Embedding-8B. Instability is the mean pairwise cosine distance among the thirty summaries. Within every corpus instability falls with n and tracks G (Spearman $+0.40$ to $+1.00$, seven of eight at $+0.70$ or above). Across corpora at a fixed n, Spearman of instability with G is $+0.14$, $+0.12$, $+0.12$, $+0.07$ and -0.17 at $n = 25$ to 400, with bootstrap bands that include zero. TIFU illustrates why: its G is near 1 at the shared radius and its summaries are the most stable, because the model summarizes it as stories of mishap whichever thirty stories it is given.

E A bound for functions that are stable at scale r

For a document function f with range 1 that takes the same value on documents within r of each other, the estimate of the corpus mean of f formed by giving each corpus document the value of its nearest sampled document errs by at most $v(r) + M(r)$, where $v(r)$ is the share of covered documents whose nearest sampled document disagrees with them on f. On fifty cluster indicator functions over four corpora, six sample sizes and six radii, the bound held in 144 of 144 cells and was loose for proportions (median 22 times the realised error), where the plain sample mean already errs by 0.003 to 0.07.

F Prompts

Summary. "Below are $\{n\}$ documents sampled from a larger collection. Write a summary of the collection in one paragraph of at most 150 words. Cover the main themes in proportion to how often they appear, name notable minority themes briefly, and do not mention individual document numbers."

Theme naming. "The $\{m\}$ documents below were grouped together because they are about the same theme. Give the theme a short name (at most 8 words) and a one-sentence definition that would let a reader decide whether a new document belongs to it. Answer as JSON with keys name and definition."

Enumeration. "Below are $\{n\}$ documents sampled from a larger collection, followed by a list of $\{K\}$ themes found in the full collection. Decide which themes are present among these $\{n\}$ documents. A theme is present only if you can point to one document that is primarily about it; in a small sample most themes will be absent, and similar-sounding themes are distinct, so pick the single best theme for each document. Answer with one line per present theme in the form theme_id: document_number."

Same content. "Two documents from the same collection are shown. Judge their relation. SAME: they are about the same specific content (the same claim, question, event or story), so a reader who has read one would learn little new from the other. TOPIC: they are about the same general topic but different specific content. UNRELATED: they are about different topics."